

Animal evolution

The first animals were microscopic and had very simple bodies. Over millions of years, their descendants evolved into new species, with new shapes and ways of life. Evolution has produced an astonishing variety of animals, from tiny insects to colossal dinosaurs.

Many species have become extinct, but others are very much alive.

Traces of the past

Scientists study extinct animals by looking at fossils. These show what prehistoric animals looked like and even how they behaved. Fossils form when buried remains slowly turn to stone.

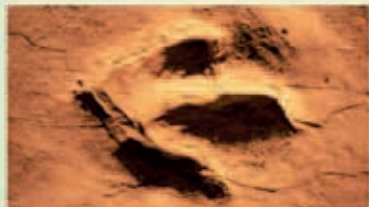
Fossil bodies

Hard-bodied animals make good fossils. This fossil belongs to an extinct trilobite, which was buried by sediment on the seabed.



Trace fossils

Some fossils preserve marks that animals have left behind. This three-toed footprint was left by a carnivorous dinosaur, or theropod.



Fossil skeletons

Bone often fossilizes, although complete fossil skeletons are a lucky find. This one belongs to a crested dinosaur called *Parasaurolophus*.



Frozen bodies

Entire animals are sometimes preserved in permanently frozen ground. This baby mammoth was found in Siberia.



Stuck in amber

This insect was trapped by sticky resin oozing from a tree. The resin slowly turned to fossil amber with the insect still inside.



Grasping, or prehensile, tail anchors body while animal attacks prey

Grasping feet are an adaptation for tree-climbing lifestyle

Evolution in action

By studying living animals and fossils, scientists attempt to chart the path that evolution has followed. This family tree, or cladogram, shows how primates have evolved. The black points on the baseline mark new adaptations that have built up by evolution over millions of years. These adaptations are shared by all the animals farther along the tree.



Tarsiers